ERA V3

This 30-session course is designed to transform students into full-stack AI engineers, proficient in both the development and deployment of AI models. The curriculum integrates foundational knowledge with advanced topics, covering neural networks, transformers, LLMs, and GenAI models. Practical skills are emphasized through continuous integration of MLOps practices, frontend and backend development, and deployment strategies.

A key differentiator of this course is the heavy use of modern coding tools like **Cursor** and **Claude dev**, which significantly enhance coding efficiency, debugging, and experimentation. These tools are a boon for accelerating learning and development, allowing students to focus on building full-stack AI solutions more effectively. By embracing these technologies, students will not only become more productive but also more confident in tackling complex AI and engineering challenges. These tools empower learners to create, debug, and experiment at a faster pace, making the journey to becoming full-stack AI engineers smoother and more accessible.

By introducing LLMs early in the course (Session 9), students have ample time to delve into advanced language models and their applications. MLOps, CI/CD, and deployment practices are interwoven throughout the sessions, ensuring students build these critical skills progressively.

The course culminates in a capstone project, allowing students to apply their knowledge to real-world problems and demonstrate their capabilities as full-stack AI engineers. The inclusion of emerging topics and future trends prepares students to adapt to the rapidly evolving field of AI.

Key Features:

- **Hands-On Learning:** Practical assignments accompany each session, reinforcing theoretical concepts through implementation.
- **Empowering Modern Tools:** Modern AI development tools like Cursor and Clade dev streamline workflows, allowing students to build, debug, and experiment faster.
- **Parallel Skill Development:** Frontend, backend, MLOps, and AI modeling skills are developed concurrently.
- **Industry Relevance:** The curriculum reflects current industry practices and technologies, ensuring students are job-ready.
- **Comprehensive Coverage:** Topics range from foundational principles to advanced techniques in AI and machine learning.
- **Focus on Deployment:** Emphasis on deploying models effectively across various platforms, including cloud and mobile devices.

This course ensures that students not only understand modern AI and deep learning concepts but are also equipped with the cutting-edge tools and skills needed to implement and deploy AI solutions effectively in real-world scenarios.

Session 1: Introduction to AI, Neural Networks, and Development Tools

- **Fundamentals of AI and Neural Networks:** Introduction to core AI concepts, types of neural networks, and their applications.

- **Course Structure and Expectations:** Overview of the course syllabus, objectives, and assessment methods.

- **Development Environment Setup:** Installing Python, PyTorch, and essential libraries.

- **Introduction to Modern Coding AI Tools:** Overview of tools like Cursor to enhance coding efficiency.

- **Introduction to Frontend and Backend Concepts:** Brief discussion to prepare for integrating AI models into applications.

Session 2: Python Essentials, Version Control, and Web Development Basics

- **Python Programming for AI:** Essential Python syntax and data structures relevant to AI programming.
- **Version Control with Git and GitHub:** Basic commands, branching, merging, and collaboration workflows.
- **Web Development Introduction:** Basics of HTML, CSS, and JavaScript to create simple web interfaces.
- **Setting Up a Simple Web Server:** Hosting applications locally to interact with AI models.

Session 3: Data Representation, Preprocessing, and UI Integration

- **Understanding Data Types:** In-depth look at how images, text, and audio data are represented in computational systems.
- **Data Preprocessing Techniques:** Cleaning, normalizing, and preparing data for model training.
- **Data Augmentation:** Techniques to artificially expand visual, textual, or aural datasets.
- **Integrating Data with Frontend Interfaces:** Displaying data on web interfaces for better understanding and user interaction.
- **Building Simple Web Apps for Data Visualization:** Creating basic applications to visualize datasets.

Session 4: PyTorch Fundamentals and Simple Neural Networks

- **Introduction to PyTorch and Tensors:** Understanding tensors, tensor operations, and PyTorch basics.
- **AutoGrad and Computational Graphs:** Mechanism of automatic differentiation in PyTorch.
- **Building Simple Neural Networks:** Constructing basic neural networks using PyTorch.
- **Implementing Training Loops:** Writing loops for training and validating models.

- **Deploying Models via Web Interfaces:** Hosting the neural network on a web server for user interaction.

Session 5: Introduction to Deployment, CI/CD, and MLOps Basics

- **Basics of Model Deployment:** Steps involved in deploying AI models to production.
- **Introduction to CI/CD Pipelines:** Automating testing and deployment using tools like GitHub Actions.
- **Introduction to Docker:** Basics of containerization for consistent deployment environments.
- **Deploying Neural Network Models in Containers:** Packaging applications for scalability and reliability.

Session 6: Convolutional Neural Networks and Training on Cloud (CNNs)

- **Basics of CNNs:** Understanding convolution operations, filters, feature maps, and receptive fields.
- **Implementing CNNs in PyTorch:** Building and training CNN models on image datasets.
- **Training CNNs:** Techniques for effective training and avoiding overfitting.
- **Deploying CNN Models with Frontend Interfaces:** Integrating trained CNN models into web applications for image classification tasks.

Session 7: In-depth Coding Practice - CNNs

- **Hands-On Practice with CNNs:** Extensive coding session focused on deepening understanding of CNN implementation.
- **Advanced CNN Architectures:** Exploring more complex CNN structures like VGG and Inception networks.
- **Data Augmentation for CNNs:** Applying data augmentation techniques to improve CNN performance.

- **Model Evaluation and Debugging:** Practical examples on how to evaluate CNNs' performance, debug issues, and fine-tune models.

Session 8: Introduction to Transformers and Attention Mechanisms

- **Understanding Attention Mechanisms:** Self-attention and its significance in deep learning.
- **Transformer Architecture:** Key components like multi-head attention and positional encoding.
- **Implementing Basic Transformers:** Building a simple transformer model in PyTorch.
- **Comparison with CNNs:** Analyzing differences and use cases for each architecture.

Session 9: Advanced Neural Network Architectures

- **Deep Residual Networks (ResNet):** Understanding residual connections and their benefits.
- **Vision Transformers (ViT):** Applying transformers to vision tasks.
- **Implementing Advanced Models:** Building and training ResNet and ViT models.
- **Deploying Advanced Models:** Handling the challenges of deploying complex architectures.

Session 10: Introduction to Large Language Models (LLMs)

- **Architecture of LLMs:** Exploring models like GPT and BERT.
- **Applications in NLP Tasks:** Text generation, sentiment analysis, and question answering.
- **Implementing Simple LLMs:** Building and training LLMs for basic tasks.
- **Deploying LLMs with Frontend Interfaces:** Creating chatbots and language assistants.

Session 11: Data Augmentation and Preprocessing

- **Image Augmentation Techniques:** Explore flipping, rotating, scaling, and adding noise to expand image datasets.
- **Text Preprocessing and Tokenization:** Learn text cleaning, tokenization, and methods like stemming and lemmatization.
- **Handling Different Data Types:** Preprocess image, text, and audio data for neural network compatibility.
- **Speeding Up Preprocessing:** Implement efficient data pipelines using batching, GPU-acceleration, and parallel processing.

Session 12: Advanced CI/CD, MLOps, and Deployment Practices

- **Advanced CI/CD Pipelines:** Implementing robust pipelines with testing and monitoring.
- **Automating Deployments:** Continuous integration and continuous deployment strategies.
- **Using Docker and Kubernetes:** Scaling applications with container orchestration.
- **Best Practices in MLOps:** Model versioning, reproducibility, and governance.

Session 13: Frontend Development for AI Applications

- **Building Responsive Web Interfaces:** Using modern frontend frameworks like React.
- **Integrating AI Models with Frontend Applications:** Connecting backend AI services with frontend interfaces for seamless user experiences.
- **Enhancing User Experience (UX):** Designing intuitive interfaces for AI applications.
- **Security Considerations:** Implementing authentication and secure communication between frontend and backend.

Session 14: Optimization Techniques and Efficient Training

- **Optimization Algorithms:** Understanding SGD, Adam, and other optimizers.
- **Learning Rate Schedules:** Techniques like One Cycle Policy for faster convergence.
- **Mixed-Precision Training:** Using FP16 and BF16 for performance gains.
- **Distributed Training Basics:** Introduction to training across multiple GPUs.

Session 15: Visualization Techniques for CNNs and Transformers

- **Visualizing CNN Models:** Techniques like Class Activation Maps (CAM) and Grad-CAM to interpret CNN decisions.
- **Visualizing Transformers:** Understanding and visualizing attention mechanisms in Transformers.
- **Interpreting Model Decisions:** Analyzing what models learn and how they make predictions.
- **Incorporating Visualization into Applications:** Displaying model insights within web interfaces for transparency.

Session 16: Generative Models: VAEs and GANs

- **Understanding Generative Models:** Concepts of latent spaces and data distribution.
- **Implementing VAEs:** Building variational autoencoders for data generation.
- **Implementing GANs:** Creating generative adversarial networks for realistic outputs.
- **Applications:** Style transfer, image synthesis, and data augmentation.

Session 17: Stable Diffusion and Advanced Generative Techniques

- **Introduction to Stable Diffusion Models:** Understanding diffusion processes in AI.
- **Implementing Stable Diffusion:** Techniques for high-quality image generation.
- **Advanced Techniques:** Inpainting, outpainting, and style transfer applications.
- **Deploying Generative Models:** Integrating into web interfaces for user interaction.

Session 18: LLM Fine-Tuning and Optimization

- **Fine-Tuning Techniques:** Supervised Fine-Tuning (SFT) and Low-Rank Adaptation (LoRA).
- **Optimization Strategies:** Parameter-efficient methods, quantization, and pruning.
- **Tools for Optimization:** Utilizing BitsAndBytes, FlashAttention, and DeepSpeed.
- **Deploying Optimized LLMs:** Balancing performance with resource constraints.

Session 19: LLM Inference and Serving

- **Inference Optimization:** Techniques like kv-cache for efficient inference.
- **Model Quantization:** Reducing model size for faster deployment.
- **Serving LLMs Efficiently:** Best practices for API and web app deployment.
- **Real-World Applications:** Chatbots, virtual assistants, and language services.

Session 20: In-depth Coding Practice - LLMs

- **Hands-On Practice with LLMs:** Intensive coding session for deepening understanding of language models.
- **Fine-Tuning Pre-trained LLMs:** Exploring strategies for fine-tuning LLMs on specific tasks or datasets.
- **LLM Evaluation and Debugging:** Practical examples on how to evaluate and improve LLM performance.
- **Optimizing Inference:** Techniques for faster LLM deployment in production settings.

Session 21: LLM Agents and AI Assistants

- **Creating LLM Agents:** Building agents that perform complex, goal-oriented tasks.
- **Using Frameworks like LangChain:** Simplifying the development of conversational AI.
- **Implementing AI Assistants:** Designing assistants for customer service, information retrieval, etc.
- **Deploying AI Assistants with Web Interfaces:** Enhancing accessibility for end-users.

Session 22: Multi-modal AI Models

- **Combining Vision and Language Models:** Techniques for integrating different data modalities.
- **Implementing Chat Systems with Image Input:** Building applications that understand both text and images.
- **Deploying Multi-modal Applications:** Addressing challenges in serving complex models.
- **Use Cases and Applications:** Real-world examples like visual question answering.

Session 23: Retrieval-Augmented Generation (RAG)

- **Effective Prompting Techniques:** Crafting prompts to guide LLM responses.
- **Introduction to LlamaIndex:** Tools for indexing and retrieving relevant information.
- **Building RAG Systems:** Combining retrieval mechanisms with generative models.
- **Deploying RAG Applications:** Creating systems like ChatWithPDF for enhanced utility.

Session 24: Advanced MLOps and Data Engineering

- **Model Versioning and Experiment Tracking:** Using MLflow and DVC for reproducibility.
- **Monitoring and Logging:** Setting up systems to track model performance.
- **Scaling AI Applications:** Techniques for handling increased load and data volume.
- **Data Pipelines and Workflow Automation:** Using tools like Apache Airflow.

Session 25: Edge AI and Mobile Deployment

- **Deploying Models on Mobile Devices:** Techniques for Android and iOS platforms.
- **Frameworks for Mobile AI:** Utilizing TensorFlow Lite and PyTorch Mobile.
- **Optimizing for Edge Deployment:** Reducing model size and improving efficiency.
- **Applications:** Real-time inference and Internet of Things (IoT) devices.

Session 26: Cloud Computing and Scalable AI

- **Deploying on Cloud Platforms:** Utilizing AWS, Azure, or Google Cloud services.
- **Cloud Services for AI:** Leveraging tools like AWS SageMaker and Azure ML.
- **Scaling with Cloud Infrastructure:** Auto-scaling and load balancing strategies.
- **Cost Optimization:** Managing resources to minimize expenses.

Session 27: In-depth Coding Practice - Scaling Up

- **Hands-On Practice with Scaling Models:** A session dedicated to scaling models using cloud and edge deployment techniques.
- **Optimizing Model Performance for Large-Scale Applications:** Implementing model partitioning, and distributed inference.
- **Challenges in Scaling:** Handling network latency, server load, and monitoring.
- **Scaling Up Model Serving:** Implementing solutions for high-performance and cost-effective serving.

Session 28: Reinforcement Learning Fundamentals

- **Basics of Reinforcement Learning (RL):** Understanding agents, environments, and rewards.
- **Q-learning and Policy Gradients:** Fundamental algorithms in RL.
- **Training Simple RL Agents:** Implementing agents in environments like CartPole.
- **Challenges in RL:** Addressing issues like exploration vs. exploitation.

Session 29: End-to-End Project Deployment - A Hands-On

- **Overview of End-to-End AI Applications:** Understanding the full pipeline from data collection to deployment.
- **Data Collection and Model Training:** Preprocessing data, training models, and preparing them for production.
- **Building Frontend and Backend:** Integrating the model with a web interface and setting up backend services.
- **Deployment Strategies:** Deploying applications using Docker and cloud services.
- **Monitoring and Scaling:** Managing performance, scaling, and security considerations.
- **Hands-On Practice:** Students will deploy a complete AI application with a frontend and backend.

Session 30: Capstone Project Work

- **Project Planning and Development:** Students begin comprehensive projects that integrate course concepts.
- **Mentorship and Guidance:** Instructors provide support and feedback.
- **Peer Collaboration:** Encouraging teamwork and knowledge sharing.
- **Preparation for Presentation:** Structuring the project for effective communication.